

Advanced (Habanero)

Q1. Solve the inequality $2x + 5 > 11$

- (A) $x > 3$
- (B) $x < 3$
- (C) $x \geq 3$
- (D) $x \leq 3$
- (E) $x = 3$

Q2. Solve the inequality $x - 3 < 2x + 1$

- (A) $x > -4$
- (B) $x < -4$
- (C) $x > 2$
- (D) $x < 2$
- (E) $x = 2$

Q3. Graph the solution to $x + 2 > 5$ on a number line

- (A) $x > 3$
- (B) $x < 3$
- (C) $x \geq 3$
- (D) $x \leq 3$
- (E) $x = 3$

Q4. Write the solution to $x - 2 > 4$ in interval notation

- (A) $(6, \infty)$
- (B) $(-\infty, 6)$
- (C) $[6, \infty)$
- (D) $(-\infty, 6]$
- (E) $(-6, \infty)$

Q5. Solve the inequality $3x - 2 < 2x + 5$

- (A) $x < 7$
- (B) $x > 7$
- (C) $x \geq 7$
- (D) $x \leq 7$
- (E) $x = 7$

Q6. Solve the inequality $2x + 1 > 3x - 2$

- (A) $x < 3$
- (B) $x > 3$
- (C) $x \geq 3$
- (D) $x \leq 3$
- (E) $x = 3$

Q7. Graph the solution to $2x - 1 < 5$ on a number line

- (A) $x < 3$
- (B) $x > 3$
- (C) $x \geq 3$
- (D) $x \leq 3$
- (E) $x = 3$

Q8. Write the solution to $x + 1 > 2$ in interval notation

- (A) $(1, \infty)$
- (B) $(-\infty, 1)$
- (C) $[1, \infty)$
- (D) $(-\infty, 1]$
- (E) $(-1, \infty)$

Open Ended Questions (FRQ)

Q9. Solve the inequality $x^2 - 4x - 3 > 0$ and graph the solution on a number line

Q10. Solve the inequality $2x^2 + 5x - 3 < 0$ and write the solution in interval notation

Chef's Tips

- Chef's Tips
- Read each question carefully
- Check your work